Beyond a Theory of Irresistible Desire

Hanna Pickard's new book examines the neurobiology of addiction by probing deeper than the usual questions we ask, and by deconstructing the questions themselves.

By Aaron Bornstein • May 2, 2026

SCIENCE & TECHNOLOGY PHILOSOPHY & RELIGION

What Would You Do Alone in a Cage with Nothing but Cocaine? A Philosophy of Addiction by Hanna Pickard. Princeton University Press, 2026. 352 pages.

HANNA PICKARD'S NEW BOOK What Would You Do Alone in a Cage with Nothing but Cocaine?*: A Philosophy of Addiction examines the failure of addiction science to produce treatments and to understand why people continue to use drugs when it causes them so much harm.*

I first met Hanna, who is now a professor of philosophy at Johns Hopkins University, when she was a visiting professor and I a postdoctoral researcher at Princeton in 2017. She gave a talk that flipped the study of addiction on its head. Instead of asking why "those people" use drugs, she asked "Why are we not all using drugs, all the time?" This was my first experience with her signature approach—uncovering the question begged by the question asked and recentering the humanity of the scientific subject. We later collaborated on a paper arguing that individuals' memories of their early experiences with drugs play a causal role in persistent drug use and subsequent relapse, drawing on first-person accounts and mathematical models to track the widely reported phenomenon of euphoric recall or "chasing the first high."

Hanna's new book is the culmination of her decades spent working with patients and researchers—first as a clinician in the United Kingdom's National Health Service (NHS), then in collaboration with scientists who study the behavior of addicted individuals or of animal proxies in the lab. The paradigm shift she proposes, expressed in accessible terms, is heterogeneous *and* humanistic. *Taking account of subjective experience and life circumstances, it argues that the scientific study of addiction should be as varied as the people suffering from it.*

The following interview is stitched together from a Zoom call, several voice memos and text messages, and in-person conversations during her mid-January book launch at the Baltimore

Museum of Art.

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AARON BORNSTEIN: How did you become interested in studying addiction?

HANNA PICKARD: By training, I'm an analytic philosopher, but I've always been interested in questions that brush up against the empirical world—especially about minds and ethics—and I became increasingly dissatisfied working only in the proverbial ivory tower. I remember saying at various points that if I could do it again, I would be a psychologist or psychiatrist who dabbled in philosophy of mind, as opposed to a philosopher who was always buried in a book or article about mental disorders or the sciences of the mind. Ultimately, I decided to volunteer in an NHS therapeutic community for people with personality disorders and complex needs, many of whom also have drug or behavioral addictions. I loved the work and eventually got hired as an assistant team therapist, a job I held part-time for a decade while also working part-time as an academic philosopher.

Therapeutic communities constitute distinctive clinical contexts. They upend the typical doctor-patient hierarchy, demanding instead genuine, egalitarian relationships and, true to the name, "community." In the place where I worked, we did all sorts of therapy together—psychodynamic, cognitive, dialectical, and mentalization-based; medication and diagnostic groups; psychodrama; and more—but we also just spent a lot of time cooking, eating, playing games, going on outings, and running the business of the group. It was a transformative experience for me and the foundation for much of the academic work I've gone on to do, including in my work on addiction.

Before I worked in this clinical setting, there had been people in my life who struggled with addiction, but I hadn't given serious thought to what addiction *is*. I just assumed it was a brain disease that compelled people to use drugs. This assumption, however, could not be reconciled with what I was seeing in the clinic. To give one example: Our group members would create contracts, publicly committing themselves to not using drugs and to getting support from the group if tempted to use. They would write this down on a piece of paper and sign it, together with the rest of the group. All of us also wrote messages of support.

Not everyone quit, but many people did. Some people carried their contracts with them everywhere, until they were ragged and worn. It was extraordinarily moving. A contract is, literally, a public commitment and a piece of paper. No brain disease involving compulsion could be cured by just that. It makes no sense. And so the philosopher in me was hooked. I wanted to understand what this thing called addiction is, which, on the one hand, is such a destructive force in people's lives, and yet, on the other, could in some cases be "cured" by making a commitment to yourself and people who care about you and whom you care about —and getting their support.

I remember you used to give an intentionally provocative talk. One slide listed all the "positive things that drugs do for us," and then you said something like "The question isn't why '*those people*' use drugs but 'Why aren't we *all* doing drugs *all the time*?'" This isn't the sort of question often asked in scientific settings. How did you get there?

I was trying to hammer home the point that drugs have tremendous value. We all know this if we stop and think about how and why we use drugs in our ordinary, day-to-day lives—remember, "drugs" are not just so-called "illicit" substances but also run the gamut from caffeine to alcohol to nicotine to opioids to sedatives to sleeping pills to amphetamines to khat to kratom and more. As a species, we have always used drugs. The reason is obvious—namely, their multiple and varied benefits, their *value*. But also, in using this line in my talks, I was trying to get people to stop and think about the invidious "us" versus "them" that permeates so much of our thinking about drugs and addiction—and to recognize that if *you* have a cup of coffee in the morning or a glass of wine in the evening, you yourself are using a drug.

The brain disease model explains drug use in addiction as a result of something wrong with the person's brain, rather than acknowledging that drugs have value. And this, of course, cements that invidious line between "us" and "them"—*they* have a broken brain and can't control themselves, while *we* don't and can.

Of course, there are differences between drug use in addiction and ordinary, day-to-day drug use. In the book, I argue that the crucial difference rests on whether, on balance, drugs are, or are not, *good* for a person. In addiction, drug use can come to undermine a person's well-being, which creates what I call the puzzle of addiction—the task of explaining why a person would nonetheless keep using drugs. But the crucial point is that, even when the costs ratchet up, the value of drugs is not eradicated, which is part of why people stay stuck. We therefore need to recognize that value in order to help them. Reckoning with the idea that drug use is not black-and-white but gray is part of the core project of the book.

But it's funny to be reminded of that slide! I don't know if I would use it today—or that I would *need* to use it today. I think there was less airspace for alternatives to the brain disease model a decade ago. My own views have also become more nuanced over time—less a response to what I think is wrong and more an attempt to really get at what's right. So, although there are no doubt places in the book that are provocative, I'm probably less provocative now.

Oh? What's the title of the book again?

Good point, good point!

How *did* you get away with this title?

It certainly met with resistance. Many people told me not to use it. But now, with the book's publication, everyone seems to love it. The book makes the case that understanding addiction is not just a scientific project but also a humanistic and imaginative endeavor. I directly address *you*, the reader, throughout the book—as I do in the title—to ask you to engage in this endeavor. But the title also refers to a seminal study with rats, conducted by the researchers Michael Bozarth and Roy Wise in the early 1980s in Montreal. The rats learned to press a lever to get a dose of cocaine, delivered intravenously. They were then permanently housed in an experimental chamber containing only food, water, and the lever, which they could press for as much cocaine as they wanted. As you might expect, the rats in this experiment took a lot of cocaine. They also stopped eating and drinking. Within a month, 90 percent had died. This image of a rat in a cage pressing a lever again and again—abandoning itself to cocaine at the expense of food and water—is an apt illustration of the brain disease model of compulsion. Why would any animal do this? The model's explanation is simple and elegant: drugs hijack the brain and compel "use" to the point of death. By extension, the model can—in theory—illustrate why a human might do the same.

The irony is that subsequent experiments have shown that "hijacking" is not the correct explanation for the rats' behavior. And, amazingly, you can see this for yourself if you forget about the rats for a minute and ask yourself what *you* would do, really and truly, if you were alone in a cage with nothing but cocaine. For many of us, if we're honest, the answer is that we'd take a lot of cocaine. Not because we think our brain would be hijacked but because we're able to imagine ourselves being trapped in isolation and emptiness. We imagine the boredom, the loneliness, the frustration, the suffering. We recognize that, in this cage, there is only one thing that offers any relief—cocaine.

So, by engaging in this imaginative feat, you generate in your own mind an alternative hypothesis to the brain disease model. It's not the power of drugs that made the rats press the lever for cocaine to the point of death. It's the fact that they were alone in a cage with nothing but cocaine. In other words, it's the barrenness of their environment and its impact on their psychological state that explains their behavior.

The subsequent experiments you mention in your book are also powerful. Can you describe them briefly?

These kinds of studies are called "forced-choice." They were pioneered in the early 1990s by Serge Ahmed—then in San Diego [at Scripps Research Institute], now at [the University of] Bordeaux—who had the ingenious idea of introducing a second lever into the experimental chamber, giving the rats an either-or choice between cocaine and an alternative reward, such as saccharin water. Amazingly, 90 percent of rats—including those who showed every indication of addiction—chose the saccharin water. Then, two decades later at the National Institute on Drug Abuse (NIDA), Yavin Shaham and his co-author Marco Venniro, who is not only a neuroscientist but also a wonderful artist and the illustrator of my book, switched the alternative from saccharin water to a social reward. One hundred percent of the rats—including those who showed every indication of addiction—chose a minute of playtime with another rat over drugs.

I am simplifying, but the take-home message from decades of animal and human research is that if you give rats or people meaningful alternatives to drugs, they take them. Addiction is associated with backgrounds of severe adversity, limited socioeconomic opportunities, and additional mental health conditions. In other words, being alone in a cage with nothing but cocaine is an apt metaphor for the life circumstances faced by many people with addiction.

Isn't it a little strange that we use animal models to understand how psychological and environmental factors drive these decisions? Why do we listen to rats and not people?

That is such a great question. As a society, we are, I think, very bad at tolerating complexity. We want singular and reductive explanations for phenomena that are, in truth, highly complicated. We want to think about addiction as something wrong in the brain that we can identify by studying any brain, human or rodent.

But when we start truly listening to what people with addiction have to say, we're forced to acknowledge that addiction is incredibly complicated and different from person to person. It is heterogeneous. The factors explaining one person's drug use, despite obvious and terrible costs, may not explain another's. There can be no one-size-fits-all model of addiction. Stigma around drug use doesn't help: it disinclines us, I suspect, from engaging in acts of genuine listening.

Maybe part of why the brain disease model has been so enduring is that it was supposed to be the "progressive" approach and, as such, an improvement over the moral model. And yet, as you show in the book, it still smuggles in moral stratification.

Yes, the brain disease model provides a simple, blanket excuse for using drugs. The excuse is that people with addiction can't help it—they're compelled—because drugs have hijacked their brains. But here's the thing: we only need an excuse if we're doing something morally wrong. So, the brain disease model unwittingly accepts one of the key assumptions of the moral model. This is part of why I try, in the book, to address moralism directly and interrogate that assumption. Yes, of course there are occasions when drug use is morally wrong—say, if you are getting behind the wheel of a car, or if it compromises your ability to look after your children. But there are also many instances when it doesn't involve moral wrongs.

Many scientific articles, if you read them closely, contain a range of stigmatizing background assumptions—that people with addiction are always lying, for example, or that they need to be managed and controlled rather than accorded autonomy and respect. This may be why we use rats so often in addiction research. It's easier. Certainly, I've experienced resistance to research that draws significantly on first-person testimony and attempts to integrate it with more standard scientific methods.

This idea that people with addiction might be "lying to you," or may not "know themselves," applies also to the neuroscience of pain. Scientists in both arenas try to "decode" subjective experience from brain activity to avoid relying on people to tell them what they're feeling. Can you talk about the "neurobiological craving signature" and what it implies for the science of drug use?

A neurobiological marker for "strong" cravings, it was discovered by the psychologist and neuroscientist Hedy Kober. She analyzed brain imaging data and subjective reports of "strength of craving" in response to "cues"—pictures of objects related to drugs or food—in people with and without addiction. She found that the neurobiological marker was present in response to drug cues *only* in people with addiction.

Some neuroscientists viewed her discovery as evidence of the brain disease model or, at the very least, of the fundamental neurobiological basis of addiction. But the very same neurobiological marker is *also* present in response to food cues in people without addiction. In other words, the difference between people with and without addiction is simply which cues cause the neurobiological response associated with strong cravings—drug cues or food cues. There is no difference in the nature of the neurobiological response itself.

The deconstruction of craving understood as "irresistible desire" is my favorite part of your book, and it made me wonder about the hype around GLP-1s [glucagon-like peptide-1 receptor agonists] like Ozempic.

It's important to begin by embedding the idea of an irresistible desire in the theoretical framework in which it's supposed to play an explanatory role. We spoke earlier of the puzzle of addiction—the task of explaining why a person would persist in using drugs despite obvious, terrible costs that mean it's against their own good. "Craving understood as irresistible desire" provides a simple, elegant solution to the puzzle that goes something like this: *Look, if they could behave differently, they would, but they can't* because their desire is irresistible*, so they don't.*

The rat in the cage looks as if it is driven by an irresistible desire for drugs—a compulsion to press the lever to the point of death. But if you put that same rat in a forced-choice study, it presses the lever for saccharin water or a minute of play with another rat. So, it can't be that its brain has been hijacked by drugs, creating pathological, irresistible desires. If this were correct, then it would also press the lever for drugs in a forced-choice study.

You mention Ozempic. The scientific evidence for its efficacy in addiction is very preliminary, but it looks likely that it can help some people, at least to some extent. It's currently being used off-label and without any knowledge of the long-term risks, however. It's hard not to see it as another chapter in the American search for the miracle drug that will cure our problems stemming from all other drugs. This is, to put it bluntly, magical thinking. It also traffics in the idea that addiction is a brain disease of compulsion—which it then monetizes. I would like us to be able to study Ozempic and use it safely when it is helpful but without being dragged back to the old paradigm, which has not yielded translational results and which has ample evidence against it.

I want to be very clear here: to deny that cravings are irresistible desires is not to deny that they are an important part of addiction. Not all but certainly many people with addiction describe cravings as a central part of their lived experience of addiction. This may be why Ozempic appears to help with addiction—just as it may help with *any* condition where part of the challenge is dealing with strong desires. Yet one of the most effective treatments is "contingency management," which in effect structures a person's environment to mirror forced-choice studies by offering alternative rewards on condition of abstinence. These rewards can be small—prizes, vouchers, money. A powerful version called a therapeutic workplace, pioneered by the behavioral psychologist Kenneth Silverman, offers skills training and employment. Contingency management is truly effective in helping some of the most vulnerable, hard-to-reach people, and yet it is rarely offered. This needs to change.

Presumably, it's not offered for political reasons. A lot of people would love for science to be "apolitical." But in the case of addiction science, we don't have that option. It is hard not to see a connection among the hijacked brain model, dehumanizing public discourses, and addiction policies. The connection drives ideas around "superpredators" or "meth zombies," as well as, here in California, "CARE Court"—a dystopian name for a process whereby people diagnosed with a substance use disorder can be sentenced to mandated treatment (which rarely works) and, if they don't comply, imprisonment. These policies take the paradigm to its logical conclusion. How might a shift to a humanistic and heterogeneous paradigm change this dynamic?

I wrote the book in part because I wanted to make a difference. Your question challenges the scientific community to address these dehumanizing forces instead of cleaving to a paradigm that is nothing more than dogma—not a theoretically coherent model with a solid evidence base. That science is apolitical is a pretense. We need better and more honest science. I think the brain disease model of compulsion has failed people with addiction and served only to entrench certain forms of stigma and discrimination. It is bad science. This does not mean that brain science can't illuminate many aspects of addiction. Of course it can. But we won't properly understand addiction if we continue to focus so narrowly on the brain. We need a paradigm that is interdisciplinary, integrating the natural sciences with the social sciences, the humanities, and, crucially, the voices of people who themselves live with addiction.

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